

EXPLANATION

PROVIDING explanations of the phenomena we experience is a major accomplishment of science. A theory should be good for making sense of the world, for explaining why things happen as they do, if it is good for anything at all. Scientific curiosity begins with why questions. Why does this happen? Why are things this way? Scientific explanations are answers to why questions. Explanations are an accomplishment, and they are worth working for because they enhance our understanding of the world.

Explanations, in other words, are desirable in themselves as valuable ends. You want your theories to explain things because that's what you want, just as you might buy shiny apples because shininess is what you want. Explanation is a desirable end, but it is also a means to something else. It is useful as an indicator of truth, as shininess might be an indicator of freshness and good taste. If an apple is shiny it is likely to be fresh, so even if shininess for shininess' sake didn't matter, shiny apples would be desirable for taste's sake. Similarly, it is a good thing for a theory to be explanatory, not just for explanation's sake but for truth's sake as well. A theory that explains phenomena is more likely to be true than one that doesn't. That, at least, is the claim if explanation is an external, truth-conducive virtue. Clearly, though, it cannot be quite so simple, since the caloric theory explains why things expand when they are heated, the theory of oceanic sirens explains the tides, and the theory about scraping the skin of old, dried-out rubber off by rubbing a balloon on your hair explains why the balloon then sticks to the wall. All these theories explain things, and yet they all are false. Nonetheless, there does seem to

be something to the link between explanation and truth, and the way to find out just what it is is to analyze carefully just what it is to be a scientific explanation.

The emphasis here is on explaining observable phenomena, not on explaining other theories. This latter accomplishment is surely an interesting and valuable role that theories have, but it is an internal affair and one we have already discussed. Now we turn to external features. The theoretical claim that protons have a positive electric charge, as an example of the distinction between internal and external explanation, explains what holds electrons in their atomic orbits. It cooperates with other theories to explain the structure of atoms, objects that are never manifested in perception. No one would have asked "Why don't the electrons fly out of the atom?" had there been no atomic theory. But the charge on the proton also explains why some of the vapor trails in a cloud chamber curve to the right when the chamber is situated between the poles of a magnet but none of the tracks curve at all when the magnet is far removed. This is an event in the world of manifest phenomena. It is something observed, and explaining why it happens is part of what is meant by making contact with the world and being compatible with the data. It is an external feature if anything is. The question is, Is this external feature of explaining observable phenomena a truth-conducive virtue? We answer that by dissecting the process of explanation, first as it is portrayed in the standard model, and then with some amendments.

THE STANDARD MODEL

Any claim to be describing the form of scientific explanation is best understood not as presenting the single form into which all good explanations actually fit but as outlining a paradigm, an exemplar to which most good explanations are similar. It is a core concept with variations and extensions as particular circum-

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stances warrant. It is the central idea from which actual explanations are more or less divergent and about which actual cases are centrally located.

The standard model of this central idea is called by philosophers of science the **covering-law model** of explanation. Its initial formulation is the work of Carl Hempel (listed in the Suggested Reading). It focuses on the form rather than on the contents of an explanation and thus follows the motivation of its originators to be generally applicable to all sciences. The pattern of a good explanation will be of how it is put rather than what it says, thus avoiding specifics of content which would draw in reference to one science or another. This goal of a general, formal characterization is a noble intention, but we will have to see how far it can go.

A phenomenon is explained by demonstration that it is an instance of a law of nature. This is the basis of the covering-law model; an event is explained when we articulate the law that covers it. This happens in everyday explanation as well as in science, and an everyday example is the best kind to start with. Suppose someone with the good fortune to be unfamiliar with automobiles and traffic asks why the cars are stopping before crossing the intersection. The why question solicits an explanation, and the explanation goes something like this. There is a stop sign on the corner, and everyone must stop at a stop sign. It's the law. The similarity between this example and laws of nature and scientific explanation is only of the form of presentation, citing a general law to cover a specific instance, and not at all of content. Traffic laws have an author and are imposed by us; natural laws have neither of these features.

There are two components of this and other good explanations. It specifies the relevant conditions of the particular case, that is, the properties of the event that are important to its happening as it does. In this case it is the presence of a stop sign on the corner. Furthermore, the explanation specifies the relevant law that

relates the conditions of the case to the phenomenon in question. It's the law that cars must stop where there is a stop sign. This law, in other words, covers this case. So, once the law is understood and the conditions are recognized, the event is to be expected. Had the law and conditions been known, the event could have been predicted. This kind of understanding is what is expected of a good explanation, and the major achievement is the finding of the relevant law.

This model seems to fit nicely onto a lot of important examples of scientific explanations. Anyone who has spun anything like a ball on a string or a little brother by his ankles knows that revolving objects will fly off, away from their orbital center, unless there is some active force holding them in. If the string breaks, the ball flies off. If you let go of your brother's ankles, he's in the bushes. Well, the moon is revolving around the earth at a pretty good clip. Why doesn't the moon fly off? There's no string, it's not set in a rigid, crystalline sphere, and nobody has it by the ankles. This deserves an explanation. And the explanation is simple: gravity. The moon doesn't fly off because of gravity. This compact explanation can be magnified to reveal its vital parts. The appeal to gravity is an appeal to a law of nature, namely, the law of universal gravitation, to the effect that any two massive objects exert a mutual force of attraction. The relevant conditions in the earth-moon case, therefore, are simply that both are massive bodies. Given the law and the conditions, the event of an attractive force on the moon is to be expected; the particular phenomenon is covered by the law.

This is not to say that all good scientific explanations are in this form of explicit appeal to law and conditions. Often the law or the conditions are implicit, and the claim of the covering-law model is that good scientific explanations can be put into this form or something very much like it.

Once explained, a phenomenon is to be expected. This is the contribution to understanding. This expectation does not mean

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simply that the law is subtly evocative or suggestive of the phenomenon or that it puts you in the right mood to expect the phenomenon. There is a stricter connection between the law and the event in question, one that forces the expectation and removes it from subjective control. The statement of the law and the relevant conditions must entail the statement of the event. With stop signs, for example, the two claims, “everyone must stop at a stop sign” and “we are at a stop sign,” logically entail the description of the event, “we must stop.” The law is not merely a suggestion of the event; it is a logical requirement.

With this strong connection between law and event, the covering-law model is also referred to as the **deductive-nomological model** of explanation, or simply the D-N model. The word “nomological” means to pertain to laws, and **deductive** is there to describe the strict logical connection between the premises of the explanatory argument, that is, the statement of the law and relevant conditions, and the conclusion, the description of the phenomenon to be explained. It is a necessary connection in the sense that, if the law is true and if the conditions hold, the event must occur. This is the warrant for expectation of the event. This is how the covering law covers.

Following this model, an explanation is presented as an argument with premises and conclusion. Often some of the premises are not mentioned, but a full presentation of the explanation should be able to expose these and reveal a complete deductive argument. An episode of explanation typically begins with a why question. Why does event *E* happen? Or, why does it happen in the particular way that it does? We’ve already asked why the moon stays in its orbit around the earth. Such questions initiate the scientific process of inquiry. Why is the sky blue? Why are there so few fish bones found at the sites of Greek Bronze Age settlements? Answers to these questions, that is, explanations, are presented as an argument: It’s because,

Reading the Book of Nature

if law L (or laws $L_1, L_2, \dots$) is true
and conditions $C_1, C_2, C_3, \dots$ hold,
then event E must be true.

To give an explanation of event E is to specify the laws L and the conditions C . Laws are recognizable as being general theories, whereas conditions are specific, observable features of the situation.

The moon example fits this form of presentation as an argument. Event E in this case is that there is an attractive force on the moon. It could even be a quantified statement regarding how much force (F) is needed to hold the moon in orbit. Only one law L is required, and it too can be given a quantified form since any two massive objects (with masses m_1 and m_2) separated by a distance (r) will exert a mutual force of attraction of magnitude

$$F = G \frac{m_1 m_2}{r^2}.$$

G is the gravitational constant to reconcile our units of mass and distance with our units of force. The particular conditions of this case are: C_1 , the earth is massive (specifically, its mass is m_1); C_2 , the moon is massive (m_2); and C_3 , the distance between the earth and the moon is finite (r). Together, L , C_1 , C_2 , and C_3 deductively imply E . Set up as an explicit argument,

if L
and C_1, C_2, C_3 ,
then E .

By covering-law standards, this is the paradigm form of explanation, and recognition of a good explanation is based on its closeness to this form rather than on any specifics of content.

A richer sampling of examples of explanations and apparent explanations will raise indications that this formal analysis is insufficient. (Remember that the project here is to characterize ex-

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planation properly so that we can clearly tell when a theory is explanatory and so we can see what it is about being explanatory that indicates that the theory is likely to be true.) There will be examples of arguments that follow the D-N form but are not what we regard as good explanations. They do not, that is, answer the initiating why question, nor do they further the understanding of the phenomenon.

Consider again the law of universal gravitation, but start with a different question, asked in a different context. The setting is a geology rather than an astronomy class, and the question is, Why does the earth have a mass of 6×10^{24} kilograms? Is the earth hollow, or lead-filled, or what? If all that it takes to present a good explanation is an argument of law-plus-conditions that entail the event E (the earth has mass $m_1 = 6 \times 10^{24}$ kg), then the law of gravity will do. The earth has the mass that it does because of gravity. Expanded into proper D-N form, the alleged explanation goes like this:

L : Any two massive objects attract with force $F = G \frac{m_1 m_2}{r^2}$.

C_1 : The moon has a mass (m_2).

C_2 : The earth–moon separation is r .

C_3 : There is an attractive force between earth and moon of magnitude F .

E : Therefore, the earth must have mass $m_1 = \frac{F r^2}{G m^2}$.

The form is valid. If the law is true and the conditions hold, the event must occur. This makes it seem as if an appeal to gravity and the circumstances of the moon explain the compositional fact about the mass of the earth, but that's silly. It's not a satisfying explanation at all. The lunar factors are irrelevant to the compositional characteristics of the earth, and the law of gravitation fails to address the cause of the phenomenon. It only mentions its effects.

The problem is that the formal analysis is symmetric but explanations are not. The mass of the earth causes its gravitational force and hence the attraction of the moon, so an appeal to masses works to explain a gravitational effect. Gravity, on the other hand, does not cause mass, and so the law of gravity does not explain the existence or magnitude of masses. But the logical form of D-N explanation is blind to this asymmetry. It licenses as good explanations such backward episodes as, Why is there a stop sign on this corner? Because we stopped here. It fits the pattern if there is a law to the effect that the only place one must stop is at a stop sign, and the condition that we must stop. These entail the conclusion that there is a stop sign present. The sign is there because we stop. No, that's backward. We stop because the sign is there, and a proper explanation of why the sign is there must cite the city's efforts at traffic safety.

These examples indicate that, even if the deductive-nomological form is a necessary feature of a good explanation, it cannot be sufficient. There must be additional criteria, most likely dealing with factors of causal relevance. These factors must be based on an understanding of the content of the theory cited in the explanation and of other theories that indicate what is relevant about the conditions. A theory informs us, for example, that mass causes gravity but not vice versa.

Another kind of counterexample to the D-N model indicates the same thing. Back in the astronomy class, we can ask the same question about the moon's attraction to the earth. The phenomenon to be explained is *E*, the moon's attraction to the earth, and the D-N form requires only that this be shown to be an instance of a regularity. *E* must be deduced from a general law and the relevant conditions, like this: *C*, the moon has craters. (This is true.) *L*, anything with craters is attracted to the earth. (This is true too.) *L* and *C* deductively entail *E*, so this explanation, that the moon doesn't fly off because of craters, has the proper D-N form. It is not, however, a good explanation. It's not even close. That's

because craters, as far as we know, are irrelevant to forces of attraction. The generalization about craters is not even a law since it doesn't pick out a natural kind of thing and associate it with a particular effect. "Cratered thing," in other words, does not pick out a natural kind, at least not with respect to forces.

Thus the covering-law model of explanation includes a problem of distinguishing laws from accidental generalizations. This cannot be done simply on the basis of the form of the statement, that is, on its generality, since "all cratered things attract" is as general as "all massive things gravitate." The distinction must attend to the content of the claim to determine that the property that identifies the kind (for example, being massive) is relevant to the effect. Determination of what counts as a law, in other words, will depend on our theoretical background to guide the decision as to what is relevant and what is not. The theories we hold, then, influence what counts as an explanation. An evaluation of this external virtue, that is, will be influenced by an internal reference to other theories.

This need for determining relevance factors is shared by all explanations, even the stop-sign case. If, as in the beginning, the event in question is someone stopping at the intersection, the explanation, as allowed by the D-N form, could cite the following generalization: Everyone must stop when there is a sign that has an additional, rectangular sign below it that says "4-way." This is a true generalization (assuming that only stop signs, though not necessarily all stop signs, bear this addition), but it is irrelevant. It misses the real law by missing the relevant factor for why people stop. They do not stop in virtue of its being a four-way sign. They stop in virtue of its being a stop sign.

This account of the covering-law model of explanation and its problems has ignored several important issues regarding its adequacy. For example, what about cases of statistical laws in which the relevant conditions don't always bring about the phenomenon? The event of Sarah's death is explained by her having

leukemia. But not every case of leukemia is fatal, so why did it happen *this* time? And there are other variations from the standard model that haven't been and won't be developed here, variations that deserve the attention of a thorough analysis of explanation. This brief account, though, is sufficient to raise the main issue of explanation as a truth-conducive virtue of theories. If being a good explanation of phenomena is to function as a truth-conducive virtue, it must be a feature we can evaluate. What *is* a good explanation? Analysis so far indicates that this feature cannot be evaluated without reference to some theories. It must be clear that the law and the conditions are describing *relevant* factors and not just circumstantial properties like having craters or a four-way sign. Any appeal to a determination of relevance will involve reference to other theories, theories that describe the fundamental workings and causal connections that make things happen as they do. Judgments of relevance, in other words, are internal judgments, getting their authority not from experience directly but from theory.

The moral of the story is that an adequate account of scientific explanation requires more than the formal guidelines provided by the covering-law model. If the covering-law model is to characterize the basic idea of explanation, it will need some amendments.

AMENDMENTS TO THE COVERING-LAW MODEL

One suggestion is to add to the formal covering-law model a measure of unification. An explanation must reveal a unity in nature, and it must be fruitful in the sense of covering a maximum number of phenomena with a minimum number of laws. This, after all, is the accomplishment of understanding, finding the most basic pattern in the events and structure of the world. The more fruitful the explanation, the better the explanation. When there are competing explanations, as between gravity and craters as the

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explanation of the moon's attraction to the earth, or between stop sign and four-way sign as explanation of the traffic stopping at the intersection, one should always opt for the more fruitful. In this way, covering-law-plus-fruitfulness will weed out the silly and irrelevant attempts at explanation.

This new scheme avoids the previous problem of irrelevant generalizations masquerading as explanations, that is, of true generalizations which, with true conditions, entail the event in question but which do so on the basis of irrelevant features. The key is that accidental associations, as between craters and an attractive force, won't be extendable to other, similar situations. The accidental generalizations won't be projectable. They won't be fruitful and unifying as is required of genuinely explanatory laws. The alleged law relating cratered objects to an attractive force could be used to explain only a limited number of cases of orbital forces, namely, those involving cratered objects like the moon, but the law relating massive objects to the force will serve all cases of moons, planets, satellites, and comets. Even without a precise indication of how to measure fruitfulness, this case is clear. The law of gravity wins, and it is to that law that one should appeal to explain the moon's behavior. Being able to unify so many different phenomena under the same law must be indicative that the law is getting at the relevant factor, mass not craters, for the effect.

The same analysis works in agreement with intuitions about the four-way-sign case. The generalization about stopping at signs with a four-way tag will cover some cases of stopping at intersections, but the generalization about stop signs will cover all those and more. This feature of fruitfulness is taken to be a sign that the generalization is describing the operative factor in the law and hasn't simply happened on a circumstantially true association.

This appeal to unification and fruitfulness will help to avoid the silly cases of explanation that result from the overly generous

symmetry of the D-N model. Unification adds the asymmetric factor into the account and blocks the sort of role switching between condition and event that allowed the moon's gravity to explain the earth's mass. The structure of the explanation will be more fruitful one way than another, thus identifying one case as the proper explanation to the exclusion of the other. All astronomical orbital systems, for example, can be explained by the law of gravity and the specific conditions of the masses of the participating objects. But not all masses can be explained by their participation in an orbital system, that is, by the law of gravity and the specific conditions of the force of attraction. This is simply because not all masses are in orbital systems. Some are isolated in deep space. Explaining the masses of these objects cannot be in terms of the forces on orbiting companions but must invoke some other generalization, thus compromising the unity in the picture of the world. The unification shown by the law of gravitation explaining all cases of orbital forces shows that the law picks out the relevant factor to bring about the phenomena and hence that it is the appropriate explanation of this kind of event. The failure of the same law to achieve the same unification in explaining the masses of objects shows that it misses the relevant point and is inappropriate as an explanation.

This example, though, also shows that the concepts of unification and fruitfulness are not as clear as they could be. In declaring the gravity-explains-orbital-attraction case to be more fruitful and of greater unification than the gravity-explains-mass case, we were operating largely on intuition and without a clearly articulated way to measure these features. Fruitfulness demands a trade-off between many phenomena explained for a few theories cited, but what is the optimal formula? If three theories buy you a hundred explanations, is that better than two theories that buy fifty? Is it simply brute numbers of phenomena explained, or is there extra credit for a large *variety* of phenomena explained? And how can you tell one variety from another? These are the

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kinds of questions that must be answered to clarify the measurement of unification and to make it useful as a criterion of explanation.

An alternative approach to amending the covering-law model of explanation is to require that an explanation include explicitly *causal* laws or causal theories of some kind. This may actually be getting at the same idea as unification in that a more unifying explanation may indicate that the law has focused on the essential causal ingredient. Though this link has been suggested in the account of unification, it has yet to be proven.

The specific focus on the causal factor in explanation represents a change of emphasis from the standard covering-law model. The goal of explanation is no longer to show that the phenomenon is to be expected. It is rather to show what brought the phenomenon about. Instead of explaining why an event happened by saying that these sorts of things always happen this way, that is, by showing it to be an instance of a regularity, the idea now is to explain the event by identifying its specific cause. Why did the dinosaurs all die? Because of a huge meteorite. The meteorite caused their deaths. Why are there tides in the ocean? Because of the moon. The moon's gravitational force causes the tides.

The requirement of an explicitly causal explanation can prevent the problems of irrelevant explanations and the silly symmetries. Since the craters on the moon don't cause the force of attraction to the earth, and no one claims that they do, the generalization about cratered objects and attraction to the earth is no explanation of the phenomenon. Similarly, the four-way sign is not what causes drivers to stop, and so even though they stop every time there is a four-way sign, this does not explain their actions. If there is no mention of the cause, then there is no relevance as an explanation. In a similar way, since the moon's attraction doesn't cause the earth to have the particular mass that it does, the moon's attraction cannot be used to explain the earth's mass, and this attempted symmetry in the argument is blocked. Stopping at

an intersection, furthermore, does not cause the placing of a stop sign, so the analysis catches up with intuition and rules, as we knew all along, that the fact that someone stops doesn't explain the fact that there is a sign. Since the causal connection is asymmetric, causal explanations are, as they should be, asymmetric.

Reference to causes looks good until we realize how difficult it can be to distinguish a genuinely causal account from an account that is not causal. How, for example, could we recognize the causal from the noncausal theories in the previous examples? That is, how do we know that craters don't cause attractive forces but masses do? The decision is founded on our theoretical background, that is, on our theories about the mechanics of gravitation. Recognition of a causal explanation, in other words, can only be done in a theoretical setting. It is clear that this characterization of explanation is no longer dependent only on the form of presentation. Different theoretical preparation will lead to different rulings on what is or is not a causal factor and hence different rulings on what is or is not really an explanation. Again, different theories in place will yield different scores on this external virtue of explanation. It is not simply a case of comparing a theory with the facts. Rather, it is a case of, under the influence of several theories, comparing a theoretical claim with the observed phenomena. In this sense, theories never confront the observed world alone but only in groups.

The source of the difficulty in identifying genuinely causal theories is that correlations between events, no matter how regular and frequent, do not necessarily report a causal connection between the events. There is a difference between two events or properties occurring together with high regularity and one event or property causing the other. Whenever the sun is high in the sky the hour hand on my clock points up high as well. But neither event is the cause of the other, though they are observed to be closely correlated. The question is, How do we tell the differ-

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ence between mere constant conjunction like this and actual causal connection, when constant conjunction is all we can detect? It is exactly this difficulty that the tobacco industry uses as a foil against arguments that smoking causes cancer. Even if there is a strong statistical correlation between incidents of smoking and incidents of lung cancer, this does not by itself support the conclusion that the smoking is the cause of the cancer. The answer is ambiguous as to which event causes the other, if there is any causal connection at all, nor does it rule out the possibility of there being a common cause, perhaps stress and anxiety, which brings about both events. This is important because if an unmeasured third condition causes both, then not smoking will not be an effective way of preventing cancer, just as resetting the hour hand of my clock will not be an effective way of moving the sun in the sky.

The moral of this story is by now familiar. Determination of causal theories, and determination of proper explanation, is neither purely formal nor purely from observational evidence. It is guided by theory, influenced, that is, by internal concerns.

EXPLANATION AND TRUTH

There is no disputing the claim that explanation is a good thing in itself. It adds to our understanding of the world, and in making the phenomena comprehensible it satisfies our scientific curiosity and helps us deal with the world we inhabit by knowing what to expect from it. It is valuable in this sense whether it is merely describing regularities or specifying actual causal relations. But does successful explanation also indicate that the explanatory theory is true? Is explanation a truth-conducive virtue?

Successful explanation certainly doesn't guarantee that the explaining theory is true. The fact that a theory can explain phenomena does not deductively prove that the theory is true. What

then is deductive about the deductive-nomological form of explanation? It is the entailment from law (and conditions) to the event. *If* the explaining theory is true (and the conditions hold), then the event in question must happen. *Assuming* the theory is true, the phenomenon is to be expected. This is clearly not a deductive proof that the theory is true, and to argue as if it is to commit a classic logical fallacy. An argument of the form “the law (and conditions) entails the event” and “the event is real” so “the law must be true” is a fallacious argument. The law being true is a sufficient explanation for the observed event, but it is not necessary. There may be other ways to explain it, and these have not been ruled out. The alleged deductive proof of the theory is just as flawed as arguing that whenever the day is the fourth of July it must be summertime, and it is summertime so it must be the fourth of July. Crazy. It could be August. The form of this argument is identical to the form of a deductive argument that a theory that explains an observed phenomenon is true. It is a form that is always fallacious.

Explanation is not a deductive proof that the explanatory theory is true, but it might contribute to an **inductive** proof. The difference between deductive and inductive proofs is exactly the difference between certainty and likelihood. The suggestion then is that successful explanation increases the likelihood that the explanatory theory is true. This would make explanation a truth-conducive virtue, if it can be demonstrated.

Does explanation in the mold of the standard covering-law model indicate that the explanatory theory is likely to be true? Recall that the accomplishment of this form of explanation is to show that the phenomenon was to be expected. A good explanation makes good sense out of what had been puzzling. It is, as C. S. Peirce put it, “feeling the key turn in the lock.” All of these criteria of good explanation, this feeling, things making sense, expectation, are psychological factors that depend on a person’s interests, antecedent beliefs, and psychological preparation. None

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of them really depends on the truth. That satisfying feeling of understanding can come from a false belief as easily as from a true belief.

A theory can be explanatory, in other words, without being true. For example, the theory that my neighbor is stealing my Sunday paper explains why the paper is not on the porch as it used to be. It all makes sense now, and it's to be expected that the newspaper won't be there. It all makes sense, but it's false. Nonetheless, if I continue to think *as if* the theory is true, I continue to make sense of the phenomena in my world. I know not to go down to look for the paper. I know I have to go buy one at the market. Now that I understand the phenomenon I can deal with it. The explanation can fully serve its purpose even though its pivotal theory is false. The same is true of more interesting examples, that explanations can succeed in making sense of phenomena, in giving us the feeling of understanding and expectation, whether they are true or false. Thinking *as if* the stuff around us is made of atoms helps us understand why the stuff behaves as it does. There's no doubt that the world is *as if* there are atoms and *as if* the guy next door pinches the paper, but it is a separate issue altogether as to whether the world really *is* this way. The psychological function of explanation, turning the key in the lock, can be accomplished without the theory being true. Truth and explanation in this sense seem to be independent issues.

Things are different if the causal factor is added to the model of explanation. A theory can explain an event, in the sense of showing that the event is to be expected, even if the theory is false, so one can accept the claim that the theory explains the event without committing oneself to believing that the theory is true. On the other hand, nothing can be the cause of an event unless that thing really exists, that is, unless the theory about it as a cause is true. Acceptance of a causal account as explanation of a phenomenon carries with it a commitment to the truth of the causal

theory. Fictions can help us make sense of observable phenomena, but fictions cannot cause real phenomena. Effective causes must be real causes.

The big idea here is to figure out whether a theory is likely to be true and to do this based on the evidence of the theory being able to explain observable phenomena. Different criteria for explanation naturally facilitate this inference to different degrees. If you know that the theory explains the phenomenon in the sense of showing that it is to be *expected*, you don't know that the theory is true. But if you know that some particular thing *caused* the phenomenon, then you also know that this thing really exists.

That's great, but how do you *know* that this particular thing is the cause of the phenomenon? A causal explanation is informative of the existence of the cause only if it is the *correct* causal account. The question then is how to tell the correct causal account from all the incorrect ones. The trouble with causal explanation as a truth-conducive virtue is not that it doesn't indicate truth (because it does indicate truth) but that it is hard to evaluate. It is hard to tell when we've got a genuinely causal explanation.

Here is a dilemma. Explanation construed as an accomplishment of expectation and understanding, that is, of psychological features, is easy to evaluate but is not indicative of truth. On the other hand, explanation construed as identification of the cause is indicative of truth but is too difficult to evaluate.

We shall have to work on this, and the work will proceed by developing a way to evaluate the goodness of a causal explanation. A causal theory will be judged in the context of other theories. The credibility of describing something as the cause of a phenomenon will be a credibility according to other theories. Thus causality and success as an explanation are evaluated by comparison to theories, and explanation as an external virtue, as a form of comparison between theory and data, is seen to have an internal component.